

BETHLEHEM, South AfricaWMO: **684610**Lat: **28.2497S**Lon: **28.3342E**Elev: **1687**StdP: **82.63**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.8	-2.3	-13.1	1.5	11.4	-11.0	1.8	9.6	9.7	11.0	8.5	11.0	1.0	50	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	29.7	15.1	28.5	15.0	27.3	15.0	18.7	24.2	18.2	23.7	17.7	23.1	4.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	15.1	20.2	16.6	14.6	19.7	16.1	14.0	19.3	60.9	24.4	58.9	23.7	57.1	23.2	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.7	6.8	-6.0	32.1	1.6	1.7	-7.2	33.3	-8.1	34.3	-9.0	35.2	-10.2	36.4
			-7.2	19.9	1.5	0.9	-8.2	20.5	-9.1	21.0	-10.0	21.5	-11.1	22.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	19.7	19.3	17.7	14.2	10.9	7.8	7.5	10.3	14.0	16.2	17.5	19.0
	DBStd	4.95	2.10	1.88	2.12	2.52	2.59	2.16	2.36	2.95	3.01	2.87	2.86	2.39
	HDD10.0	211	0	0	0	2	21	70	82	32	4	1	1	0
	HDD18.3	1588	11	10	36	124	231	315	336	249	132	77	47	20
	CDD10.0	1851	300	261	240	128	48	6	4	41	124	193	227	279
	CDD18.3	186	52	38	17	1	0	0	0	0	3	11	23	41
	CDH23.3	2149	413	295	194	38	4	0	0	15	165	276	333	414
	CDH26.7	397	84	48	21	1	0	0	0	0	19	54	73	96
(14) Wind	WSAvg	2.9	3.3	3.0	2.8	2.4	2.4	2.4	2.5	2.8	3.2	3.5	3.6	3.3
(15) Precipitation	PrecAvg	691	122	100	77	41	19	11	7	18	27	75	94	107
	PrecMax	1005	243	238	211	118	116	55	58	118	191	223	247	270
	PrecMin	436	28	8	20	0	0	0	0	0	0	15	11	23
	PrecStd	132	58	52	41	27	23	14	13	28	33	50	54	49
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	30.0	28.9	26.2	23.8	20.5	20.6	25.2	28.7	30.0	30.6	32.1
		MCWB	16.0	15.8	15.7	13.4	10.8	8.4	8.1	10.8	12.2	13.4	14.3	15.4
	2%	DB	28.9	28.2	27.0	24.5	21.7	18.6	18.9	23.0	27.0	28.2	28.8	29.0
		MCWB	16.4	16.2	15.6	13.3	10.6	8.0	7.6	10.0	11.8	13.0	14.3	15.5
	5%	DB	27.2	26.7	25.5	23.0	20.3	17.2	17.5	21.1	25.4	26.5	27.0	27.3
		MCWB	16.7	16.5	15.5	13.4	10.2	7.5	7.3	9.1	11.3	13.0	14.3	15.6
	10%	DB	25.8	25.2	24.1	21.4	18.8	15.9	16.1	19.3	23.6	24.7	25.2	25.7
		MCWB	16.8	16.6	15.6	13.1	9.9	7.2	6.9	8.5	10.7	12.7	14.3	15.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.5	18.6	16.2	13.6	10.6	10.1	12.9	15.1	16.7	18.1	19.1
		MCDB	25.3	24.4	23.5	21.5	19.1	16.6	16.1	20.7	24.2	23.2	24.2	25.2
	2%	WB	18.7	18.6	17.7	15.4	12.4	9.6	9.0	11.6	13.7	15.7	17.1	18.1
		MCDB	24.5	23.8	23.1	20.9	18.3	15.6	15.9	19.4	22.4	22.5	23.2	24.0
	5%	WB	18.2	18.0	17.0	14.7	11.5	8.7	8.2	10.6	12.6	15.0	16.4	17.5
		MCDB	23.8	23.3	22.2	20.3	17.6	14.9	15.5	18.2	21.7	21.8	22.5	23.3
	10%	WB	17.6	17.5	16.4	14.0	10.6	7.9	7.4	9.6	11.8	14.2	15.7	16.8
		MCDB	23.0	22.5	21.6	19.5	16.9	14.3	14.8	17.0	20.6	21.0	21.7	22.6
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	11.5	11.7	13.2	15.2	16.0	16.4	16.7	16.7	14.6	13.4	12.5
		MCDBR	14.6	14.5	14.8	16.1	17.8	18.1	18.7	19.3	19.6	18.4	16.7	15.3
		MCWBR	5.0	5.3	5.6	6.8	8.5	9.4	9.6	9.0	8.2	6.8	5.6	5.4
	5% WB	MCDBR	12.1	11.6	11.9	12.5	14.5	15.6	16.2	16.4	16.5	13.9	13.1	12.6
		MCWBR	4.9	4.9	5.3	5.9	7.6	9.0	8.9	8.4	7.5	6.1	5.4	5.3
(40) Clear-Sky Solar Irradiance	taub		0.359	0.355	0.342	0.304	0.273	0.257	0.257	0.319	0.358	0.388	0.345	0.350
	taud		2.380	2.395	2.403	2.486	2.555	2.565	2.544	2.358	2.254	2.215	2.380	2.396
	Ebn at Noon		983	969	950	941	932	928	942	910	917	926	990	994
	Edh at Noon		130	124	116	97	81	76	80	108	133	147	129	128
(44) All-Sky Solar Radiation	RadAvg		6.82	6.35	5.58	4.68	4.11	3.72	4.06	4.80	5.85	6.33	6.98	7.05
	RadStd		0.65	0.58	0.41	0.34	0.21	0.20	0.17	0.26	0.30	0.47	0.46	0.44

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.57	+0.71	N/A	N/A	-0.39	-0.56	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.57	+0.71	N/A	N/A	-0.39	-0.56	N/A	N/A	N/A

Nomenclature: See separate page